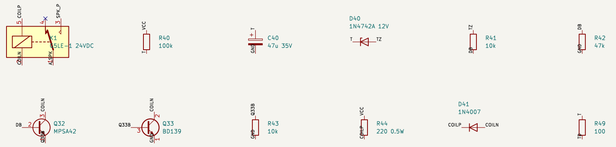
Power flags (ERC drivers for rails that enter through connectors)

Connectors



J1 rails (screw terminal). J2 signal from preamp J15. J3 speaker. J4 AC sense from one 20 VAC winding of T1.

Speaker relay with ~3.5 s turn-on delay



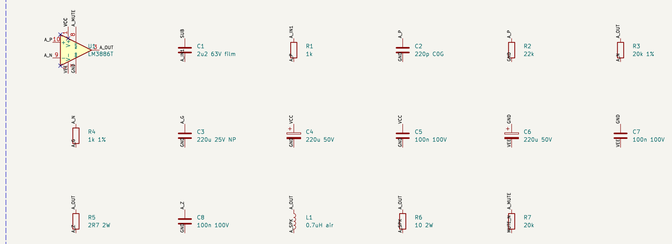
T charges via R40/C40; D40 (12 V) + Darlington Q32/Q33 pull in K1 at ~13.6 V. Anything pulling TD low drops the relay within ~5 ms.

Mute driver



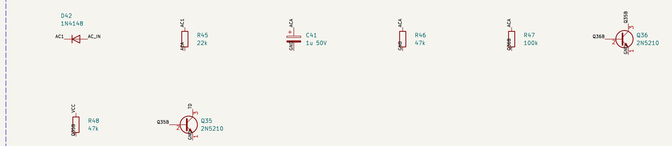
Relay energised (COILN low) → Q41 on → Q30 on → ~1.3 mA out of each MUTE pin = play. Otherwise muted.

LM3886 A (non-inverted half)



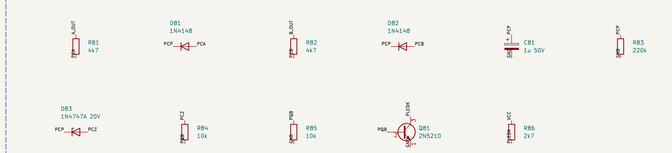
Gain 1 + 20k/1k = 21. Tab = V− : insulate from the heatsink. Gain resistors 1 % (match A to B).

AC-loss detector



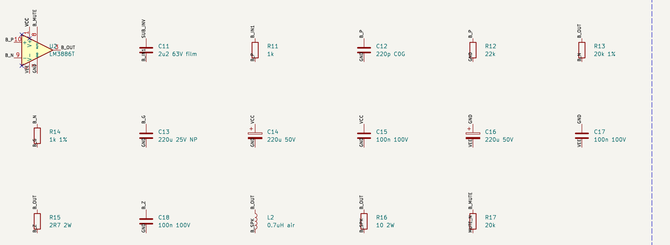
While AC is present Q36 holds Q35 off. Mains off: C41 decays in ~30 ms, Q35 dumps C40, relay opens, chips mute.

POWER clip indicator



Peak detector on both bridge halves; lights at ~21 V peak per side.

LM3886 B (inverted half)



Gain 1 + 20k/1k = 21. Tab = V− : insulate from the heatsink. Gain resistors 1 % (match A to B).

DC-fault detectors (one per output, either polarity)



Each output low-passed by 47k/47uF (bipolar). > +1.3 V turns on Q37/Q39. < −1.3 V turns on Q38/Q40; both dump TD.

J5 goes to the phones jack NC switch (opens = speaker muted). Fit a jumper if unused.
Heatsink <= 0.4 C/W (or <= 0.7 C/W + fan). LM3886T tab is V−.
Rails +/−28 V from the PSU board with heavy wire. Star ground at the PSU reservoir caps.
BRIDGED OUTPUT: neither speaker terminal is ground. Use a Speakon NL4 (1+ = SPK+, 1− = SPK−).

Sheet: /
File: poweramp.kicad_sch

Title: Bass amp power amplifier: 2 x LM3886T bridged (BTL), protection

Size: A1	Date:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Rev: 1
	Id: 1/1